DataBase Project - Phase 1

Mahla Entezari

Overview

In this project, I designed and built a database for a **Media Streaming Service** . The database stores critical information, such as

- User profiles
- Subscription details
- Payment histories
- Personalized watch lists
- ..

It also manages media assets, including movies and series with multiple episodes while tracking their storage locations and associated production companies. The system ensures that all data is efficiently organized to support user activities like

- Leaving comments
- Rating media
- Curating a watch list for future viewing
- ..

Specifications:

- Each user can subscribe to the service for a period of time. The start and end date of their subscription (including the subscription tier) must be stored.
- The users' payment history, including the related subscription and transaction details, must be stored as well.
- A media entity can be a movie or a series, and it includes a genre, a production year, an average rating score, a director, a list of actors, a list of producers, and a list of comments left by users.
- Users can leave multiple comments for a media entity but can rate each item once.
- A series consists of multiple episodes. Each movie or episode of a series must have a storage location, which stores the corresponding server and path of the media file.
- Each movie or series must belong to a production company. Each company's details, such as its name, year of establishment, and contact information, must be stored in the database.
- Each user has a 'Watch Later' list to which specific movies or episodes of a series can be added.

Requirements Analysis

1. Identifying the project's main entities and their attributes based on the specifications provided: **Entities :**

   - User

     - `UserID` -> To identify each user
     - `Username` -> For user to login
     - `Password` -> For authentication
     - `Email` -> For login and communications
     - `SubscriptionTier` -> `Free`, `Premium`, ..
     - `SubscriptionStartDate` -> Start of subscription
     - `SubscriptionEndDate` -> End of subscription

     I chose User because we want to store some details of users.

   - Payment

     - `PaymentID` -> To idetify each payment
     - `UserID` -> To show each user this payment belongs to
     - `AmountPaid` -> How much it paid
     - `SubscriptionID` -> For which subscription this payment was made

     I chose payment because we want to store some details of payments.

   - Media

     - `MediaID` -> To identify
     - `Title` -> For easy searching
     - `Type` -> `Series` or `Movie`
     - `Genre` -> Search for arbitrary genre
     - `ProductionYear`
     - `AverageRatingScore`
     - `Director`
     - `Actors`
     - `Producers`
     - `Comments`

     I chose Media because we want to store some details of Media.

   - Company

     - `CompanyID` -> To identify
     - `Name`

        - `YearOfEstablishment`
        - `PhoneNumber` -> Contact Information
        - `Location` -> Contact Information
        - `Email` -> Contact Information

      I chose Company for linking to Media.

    - Comment

        - `CommentID`
        - `UserID`
        - `MediaID`
        - `Text`

      I chose Comment to link with Users and Media.

    - Episode

        - `EpisodeID`
        - `SeriesID`
        - `Title`
        - `StorageID`

      I chose Episode to link with Series.

    - Movie

        - `MovieID`
        - `StorageID`
        - `MediaID`
        - `Title`

      I chose Movie to link with Media.

    - Series

        - `SeriesID`
        - `MediaID`
        - `Title`

      I chose Series to link with Media and Episodes.

    - StorageLocation

        - `StorageID`
        - `Server`
        - `Path`

      I chose StorageLocation to store Server and Path of the media file.

    - Rating

        - `RatingID`
        - `UserID`
        - `MediaID`
        - `Score`

   I chose Rating to store rating scores.

    - WatchLater
        - `ListID`
        - `UserID`
        - `MediaID`

   I chose WatchLater to store list of medias for watching later.

2. The relationships and constraints required to model a streaming service accurately:

Relationships

Entity1	Entity2	Type	Explanation	Constraints	Details
User	Payment	One-to-Many	A user can make multiple payments.	`UserID` in **Payment** references `UserID` in **User**.	Payments track user subscription fees and transactions over time.
User	Comment	One-to-Many	A user can leave multiple comments on media items.	`UserID` in **Comment** references `UserID` in **User**.	Comments store user feedback or reviews for movies and series.
User	Rating	One-to-	A user can rate multiple media items,	Composite unique key `(UserID, MediaID)` in	Ratings are numerical scores (e.g., 1–5) left by users for

Entity1	Entity2	Type	Explanation	Constraints	Details
		Many	but only once per item.	Rating.	media items.
User	Watch Later	Many-to-Many	A user can save multiple media items to their "Watch Later" list.	`UserID` and `MediaID` as a composite key in **WatchLater**.	This feature allows users to save movies or episodes for future viewing.
Media	Production Company	Many-to-One	Many media items can belong to a single production company.	`ProductionCompanyID` in **Media** references `CompanyID` in **ProductionCompany**.	Media items like movies and series are associated with the company that produced them.
Series	Episode	One-to-Many	A series can have multiple episodes.	`SeriesID` in **Episode** references `SeriesID` in **Series**.	Series represent collections of episodes, with each episode uniquely identified.
Movie/Episode	StorageLocation	One-to-One	Each movie or episode has one unique storage location.	`StorageID` in **Movie/Episode** references `StorageID` in **StorageLocation**.	Storage location tracks the server and file path for streaming or retrieval.
Media	Comment	One-to-Many	Each media item can have multiple comments.	`MediaID` in **Comment** references `MediaID` in **Media**.	Users leave reviews and comments on media items like movies or episodes.
Media	Rating	One-to-Many	Each media item can have multiple ratings.	`MediaID` in **Rating** references `MediaID` in **Media**.	Ratings enable calculating the average rating score for movies or series.

Attribute Constraints Table

Table	Attribute	Data Type	Constraints
User	`UserID`	`INT`	Primary Key, Not Null
	`Username`	`VARCHAR(255)`	Unique, Not Null
	`Password`	`VARCHAR(255)`	Not Null
	`Email`	`VARCHAR(255)`	Unique, Not Null
	`SubscriptionTier`	`VARCHAR(50)`	Check: Values in ('Free', 'Premium'), Default: 'Free', Not Null
	`SubscriptionStartDate`	`DATE`	Not Null
	`SubscriptionEndDate`	`DATE`	Not Null, Check: `SubscriptionStartDate` < `SubscriptionEndDate`
Payment	`PaymentID`	`INT`	Primary Key, Not Null
	`UserID`	`INT`	Foreign Key (User), Not Null
	`AmountPaid`	`DECIMAL(10,2)`	Not Null, Check: `AmountPaid > 0`
	`SubscriptionID`	`INT`	Foreign Key (Subscription), Not Null
Media	`MediaID`	`INT`	Primary Key, Not Null
	`Title`	`VARCHAR(255)`	Not Null
	`Type`	`VARCHAR(50)`	Check: Values in ('Series', 'Movie'), Not Null
	`Genre`	`VARCHAR(100)`	Not Null

Table	Attribute	Data Type	Constraints
	`ProductionYear`	`YEAR`	Not Null
	`AverageRatingScore`	`FLOAT`	Default: 0.0, Check: `0 <= AverageRatingScore <= 5`
	`Director`	`VARCHAR(255)`	Not Null
	`Actors`	`TEXT`	Not Null
	`Producers`	`TEXT`	Not Null
	`Comments`	`TEXT`	Nullable
Company	`CompanyID`	`INT`	Primary Key, Not Null
	`Name`	`VARCHAR(255)`	Not Null
	`YearOfEstablishment`	`YEAR`	Not Null
	`PhoneNumber`	`VARCHAR(20)`	Not Null
	`Location`	`VARCHAR(255)`	Not Null
	`Email`	`VARCHAR(255)`	Unique, Not Null
Comment	`CommentID`	`INT`	Primary Key, Not Null
	`UserID`	`INT`	Foreign Key (User), Not Null
	`MediaID`	`INT`	Foreign Key (Media), Not Null
	`Text`	`TEXT`	Not Null
Episode	`EpisodeID`	`INT`	Primary Key, Not Null
	`SeriesID`	`INT`	Foreign Key (Series), Not Null
	`Title`	`VARCHAR(255)`	Not Null
	`StorageID`	`INT`	Foreign Key (StorageLocation), Not Null
Movie	`MovieID`	`INT`	Primary Key, Not Null
	`StorageID`	`INT`	Foreign Key (StorageLocation), Not Null
	`MediaID`	`INT`	Foreign Key (Media), Not Null
	`Title`	`VARCHAR(255)`	Not Null
Series	`SeriesID`	`INT`	Primary Key, Not Null
	`MediaID`	`INT`	Foreign Key (Media), Unique, Not Null
	`Title`	`VARCHAR(255)`	Not Null
StorageLocation	`StorageID`	`INT`	Primary Key, Not Null
	`Server`	`VARCHAR(255)`	Not Null
	`Path`	`VARCHAR(500)`	Not Null
Rating	`RatingID`	`INT`	Primary Key, Not Null
	`UserID`	`INT`	Foreign Key (User), Not Null
	`MediaID`	`INT`	Foreign Key (Media), Not Null
	`Score`	`INT`	Not Null, Check: `1 <= Score <= 5`

Table	Attribute	Data Type	Constraints
WatchLater	`ListID`	`INT`	Primary Key, Not Null
	`UserID`	`INT`	Foreign Key (User), Not Null
	`MediaID`	`INT`	Foreign Key (Media), Not Null